

NoSQL: Operational Big Data

Data Types and NoSQL Data Models

Data Type	Data Model
CSV, TSV or web logs	Column, Document
Documents	Document
JSON	Document
Metadata catalog	Column, Document
Keyed images and documents	Key-Value
RDF, Linked data	Graph

Key-Value Stores

(key,value)

What Are They?

- NoSQL's OLTP equivalent
- Extremely simple
- Key-blob pairs, that's it
- Associative array data model
- Retrieve value given a key
 - All access is by a key

Key-Value Stores

(key,value)

Good for:

- Any single object of unstructured data
- Storing BLOBs
- Fast writes
- Web app cache
- Session Information – get all session information in a single put/get
- User profile data
- Massive multi-player on-line gaming
- Shopping carts (up until the payment transaction)
- Geo-localized processing
- Speed and when you can't be down

Key-Value Stores

(key,value)

Leading vendors:

- Riak
- Redis (a.k.a. Data Structure Server)
- Memcached DB
- Berkeley DB
- Hamster DB (embedded database)
- Amazon Dynamo DB
- Project Valdemort (open source Amazon Dynamo DB)

Document-oriented Databases



What are they?

- Key-Value Stores with added capabilities
 - Ability to nest sub-documents
- JSON/XML data models
- With Tree-Like Structure
- Encapsulated document objects
- Groups data together more naturally and logically

Document-oriented Databases



Good for:

- Semi-structured data
- Web pages
- Web traffic / E-Commerce
- Web analytics
- Log files
- User actions / behaviors
- Content Management Systems
- Full text
- Uncertain data
- Extending object-oriented approaches
- Event logging
- JSON/XML data

Document-oriented Databases



Leading vendors:

- Couchbase
- CouchDB
- MarkLogic
- MongoDB
- RavenDB
- Cloudbant (IBM SaaS)

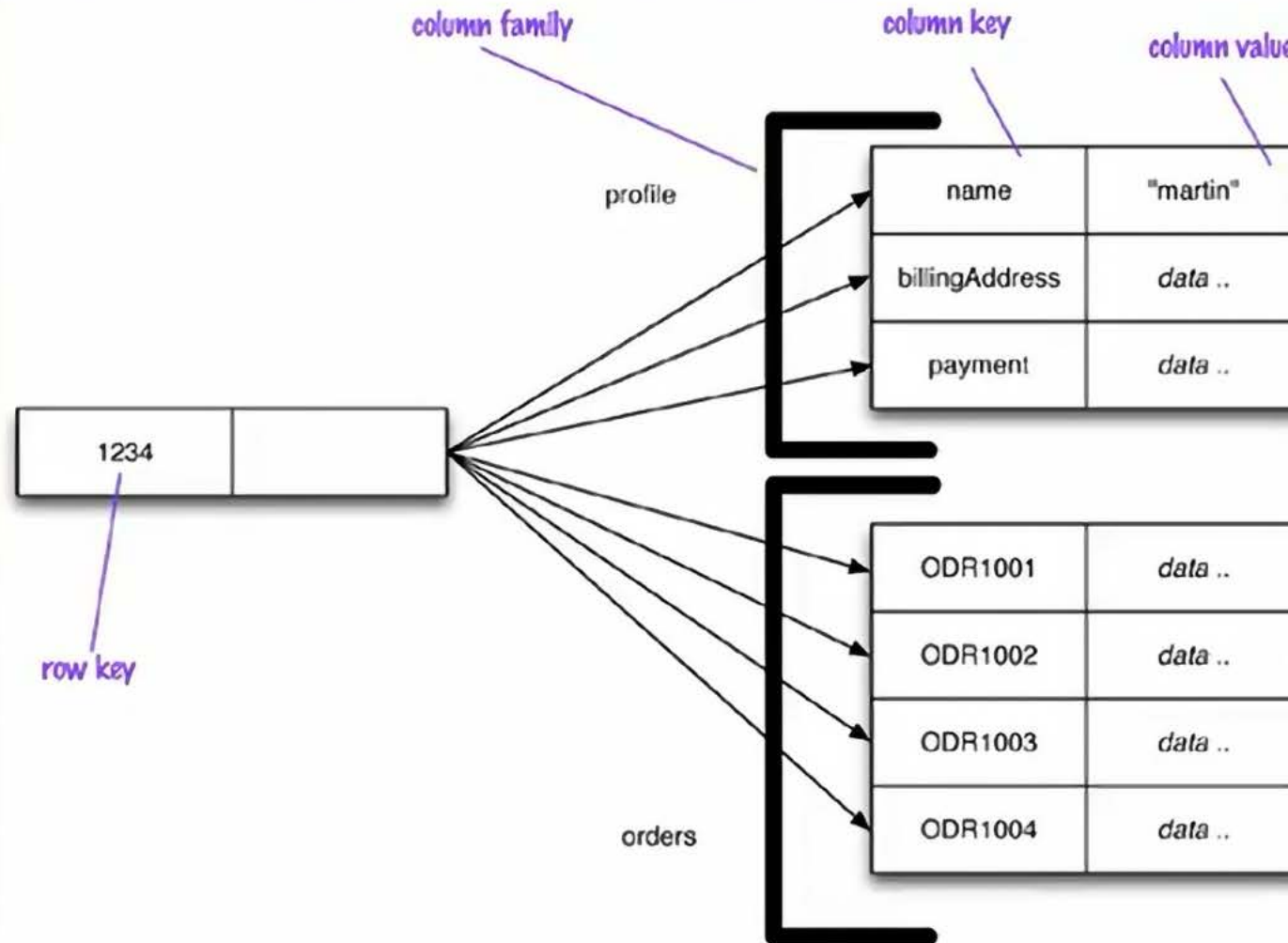
Column Stores



What are they?

- Data model:
 - A big table, with column families
 - Map-reduce for querying/processing
- Schema-lite
- No single point of failure
- Operational simplicity
- Closest NoSQL implementation to RDBMS

Column Stores Example



Column Stores



Good for:

- Large amounts of data
- Data that needs compression
- Event logging
- Content Management Systems
- Data model supports semi-structured data
- Naturally indexed (columns)
- Good at scaling out horizontally
- Time Series data
 - Weather data
 - Location data
 - Sensor data

Use Cases

- Product Catalog & Playlists
- Content Management
- Recommendation and Personalization
- Fraud Detection
- Messaging
- Internet of Things/Sensor Data
- Location-based Services

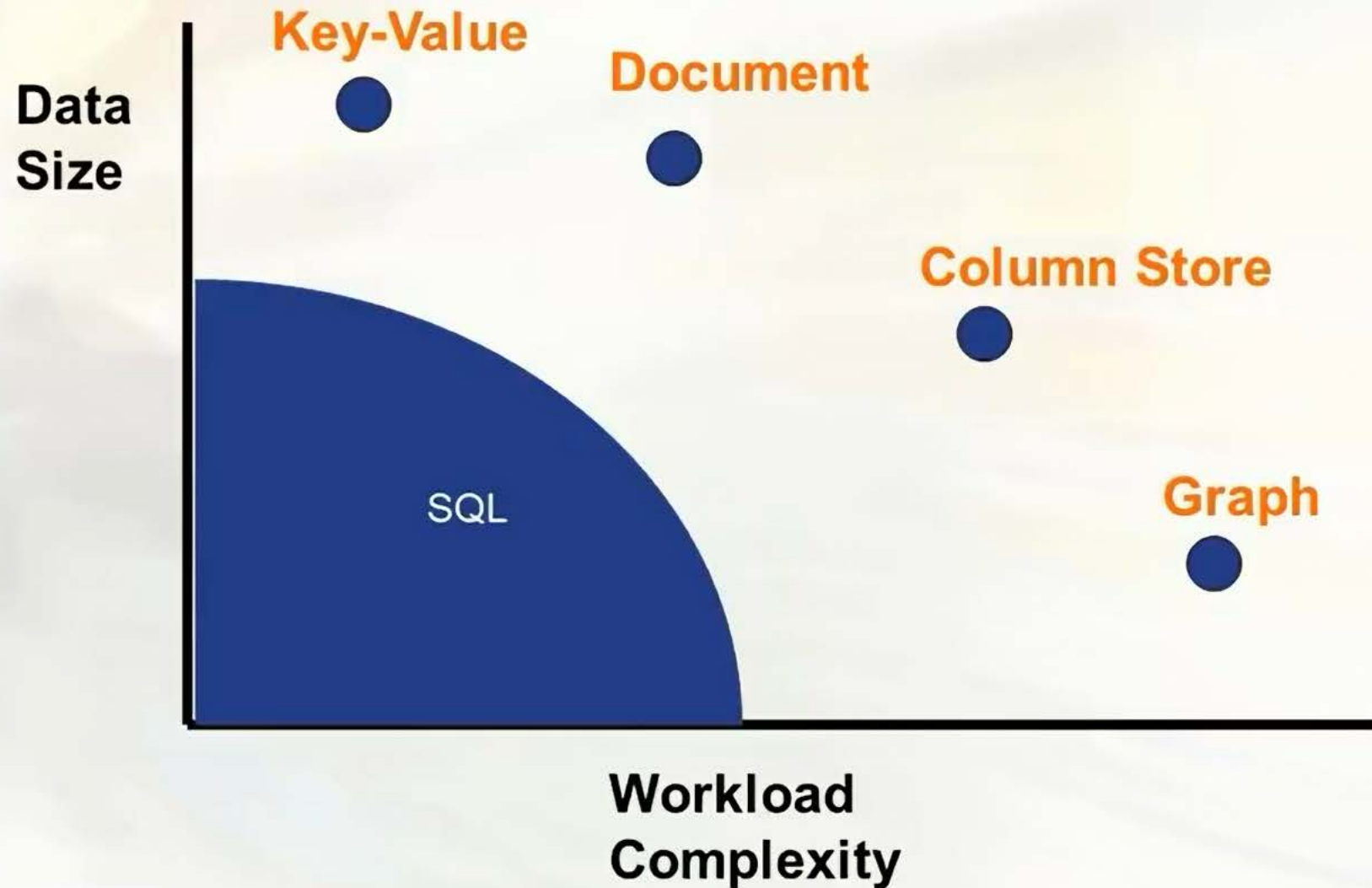
Column Stores



Leading Vendors

- Cassandra
- Hbase (Hadoop ecosystem)
- Hypertable

Operational Big Data Platform Selection



Multiple NoSQL Solutions Working Together

You could use

- Key-Value Store for Shopping Cart and Session Data
- Document Store for Consuming Completed Orders
- RDBMS for inventory (small, not served real-time), financials
- Graph Store for Customer Relationships for Marketing